

7.5) Product Analysis

Product Analysis View

Before beginning the product analysis, a dedicated SQL VIEW named **Product Analysis Data** was created to consolidate all product-related information into a single reusable analytical dataset.

The view integrates customer demographics, vision type, staff assignment, purchase history, repeat customer status, repeated customer churn status, Top 50 customer flag, invoice details, product information, quantity sold, and final product price by joining multiple normalized tables into one centralized view.

This approach eliminates the need to repeatedly write long and complex JOIN operations for every product-related analysis, making subsequent SQL queries significantly shorter, easier to maintain, and more efficient.

The view serves as the foundation for all product analytics performed in this project, including:

- Product-wise sales analysis
- Brand performance analysis
- Product type analysis
- Revenue analysis
- Quantity sold analysis
- Vision type vs product preference analysis
- Age group vs product preference analysis
- Repeat customer purchasing behaviour
- Churn customer product analysis
- Top customer product purchasing patterns

By creating this reusable analytical view, the product analysis becomes more modular, scalable, and easier to extend for future business questions without rewriting the complete data preparation process each time.

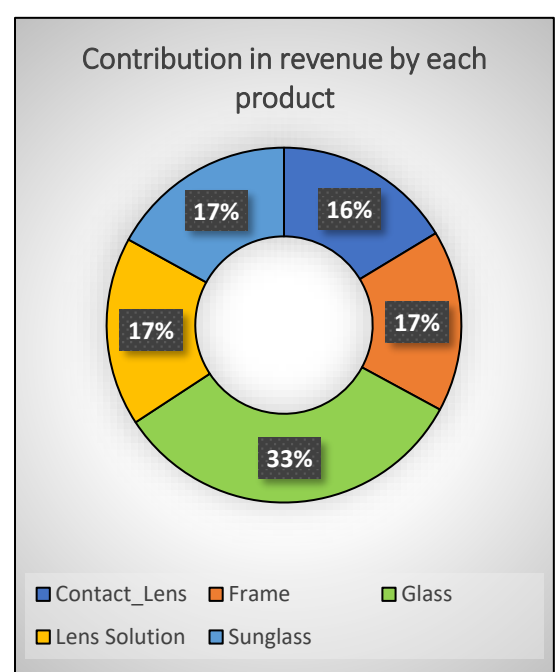
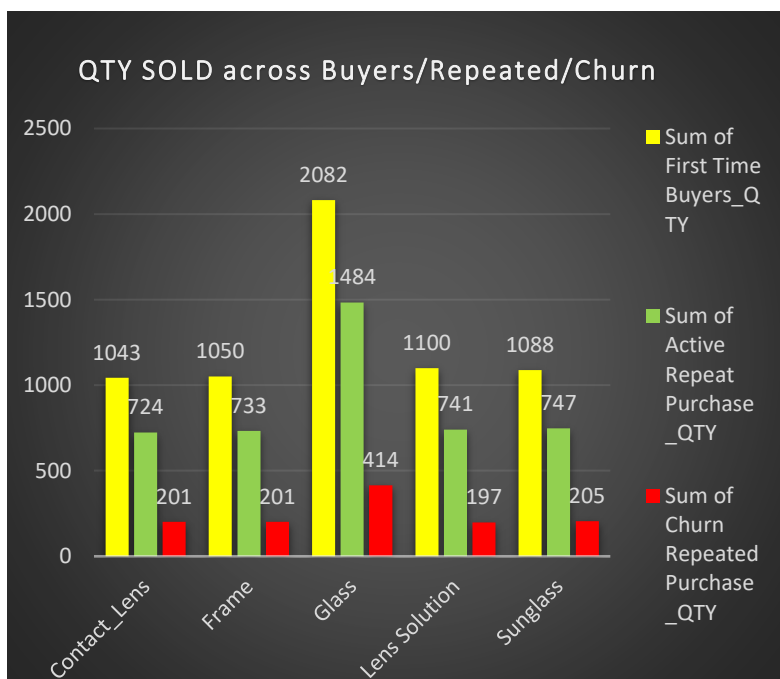
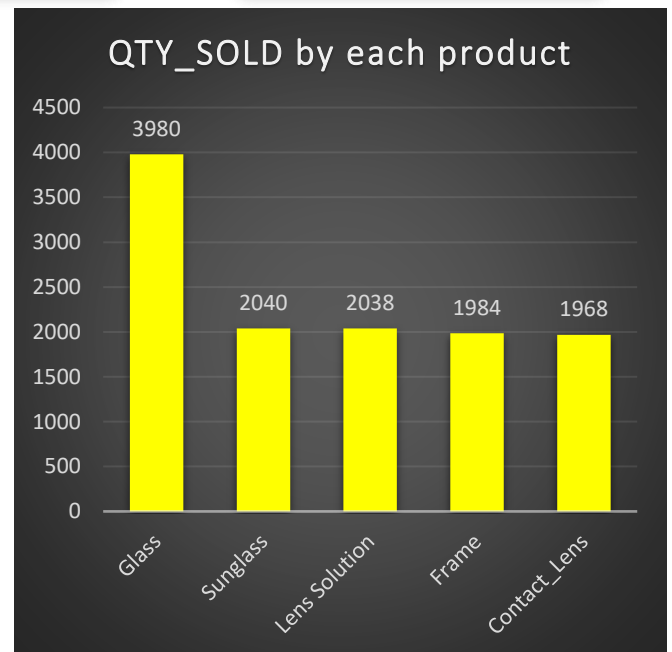
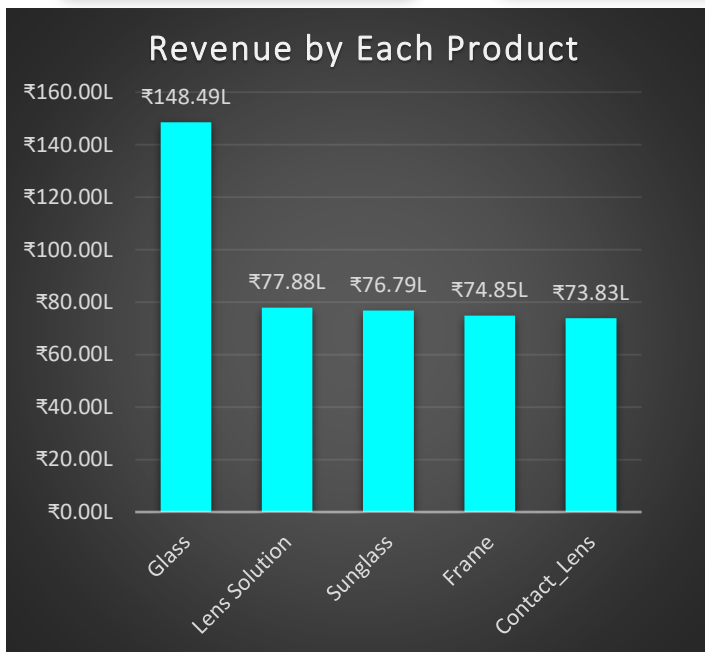
```
CREATE VIEW Product_Analysis_data AS
WITH customers AS (
SELECT DISTINCT customer_id
FROM customers_details
),
customers_detail AS (
SELECT
    c.customer_id, cd.age, cd.mobile, cd.customer_type,
    cd.assigned_staff_id, cd.referred_by_id
FROM customers as c
INNER JOIN customers_details as cd
ON cd.customer_id=c.customer_id
),
cd_vision AS (
SELECT
    cd.customer_id, cd.age, cd.mobile, cd.customer_type,
    cd.assigned_staff_id, cd.referred_by_id,
    p.vision_type
FROM customers_detail as cd
INNER JOIN prescriptions as p
ON p.customer_id=cd.customer_id
),
cdv_staff AS (
SELECT
    cdv.customer_id, cdv.age, cdv.mobile, cdv.customer_type,
    cdv.assigned_staff_id, cdv.referred_by_id,
    cdv.vision_type, s.staff_full_name
FROM cd_vision as cdv
INNER JOIN staff as s
ON s.staff_id=cdv.assigned_staff_id
),
```

```
cdvs_repeated AS (
SELECT
    cdvs.customer_id, cdvs.age, cdvs.mobile, cdvs.customer_type,
    cdvs.assigned_staff_id, cdvs.referred_by_id,
    cdvs.vision_type, cdvs.staff_full_name,
    rc.customer_id as repeated_customers
FROM cdv_staff as cdvs
LEFT JOIN repeated_customers as rc
ON rc.customer_id=cdvs.customer_id
),
cdvsr_churn AS (
SELECT
    cdvsr.customer_id, cdvsr.age, cdvsr.mobile, cdvsr.customer_type,
    cdvsr.assigned_staff_id, cdvsr.referred_by_id,
    cdvsr.vision_type, cdvsr.staff_full_name,
    cdvsr.repeated_customers,
    rcc.customer_id as repeated_churn_customers
FROM cdvs_repeated as cdvsr
LEFT JOIN repeated_churn_customers as rcc
ON cdvsr.customer_id=rcc.customer_id
),
cdvsr_top AS (
SELECT
    cdvsrct.customer_id, cdvsrct.age, cdvsrct.mobile, cdvsrct.customer_type,
    cdvsrct.assigned_staff_id, cdvsrct.referred_by_id,
    cdvsrct.vision_type, cdvsrct.staff_full_name,
    cdvsrct.repeated_customers,
    cdvsrct.repeated_churn_customers,
    cdvsrct.top_50_customers,
    tc.customer_id as top_50_customers
FROM cdvsr_churn as cdvsrct
LEFT JOIN top_50_customers as tc
ON tc.customer_id=cdvsrct.customer_id
),
```

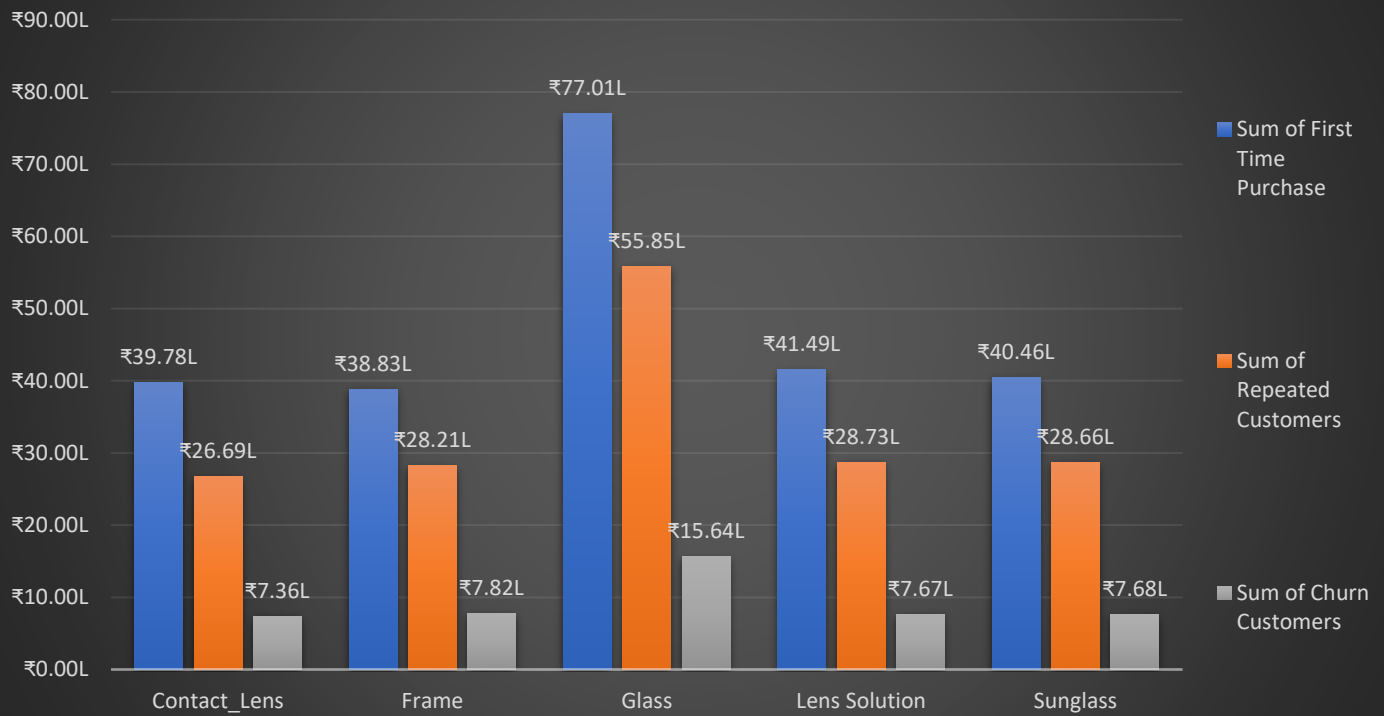
```
cdvsrct_invoices AS (
SELECT
    cdvsrct.customer_id, cdvsrct.age, cdvsrct.mobile, cdvsrct.customer_type,
    cdvsrct.assigned_staff_id, cdvsrct.referred_by_id,
    cdvsrct.vision_type, cdvsrct.staff_full_name,
    cdvsrct.repeated_customers,
    cdvsrct.repeated_churn_customers,
    cdvsrct.top_50_customers,
    i.invoice_id
FROM cdvsr_top as cdvsrct
INNER JOIN invoices as i
ON i.customer_id=cdvsrct.customer_id
),
cdvsrctli_invoice_items AS (
SELECT
    cdvsrctli.customer_id, cdvsrctli.age, cdvsrctli.mobile, cdvsrctli.customer_type,
    cdvsrctli.assigned_staff_id, cdvsrctli.referred_by_id,
    cdvsrctli.vision_type, cdvsrctli.staff_full_name,
    cdvsrctli.repeated_customers,
    cdvsrctli.repeated_churn_customers,
    cdvsrctli.top_50_customers,
    cdvsrctli.invoice_id, ii.item_id, ii.product_id, ii.brand_id, ii.product_type_id,
    ii.quantity, ii.final_product_price
FROM cdvsrct_invoices as cdvsrctli
INNER JOIN invoice_items as ii
ON ii.invoice_id=cdvsrctli.invoice_id
),
cdvsrctli_invoice AS (
SELECT
    cdvsrctli.customer_id, cdvsrctli.age, cdvsrctli.mobile, cdvsrctli.customer_type,
    cdvsrctli.assigned_staff_id, cdvsrctli.referred_by_id,
    cdvsrctli.vision_type, cdvsrctli.staff_full_name,
    cdvsrctli.repeated_customers,
    cdvsrctli.repeated_churn_customers,
    cdvsrctli.top_50_customers,
    cdvsrctli.invoice_id, cdvsrctli.item_id,
    cdvsrctli.product_id, cdvsrctli.brand_id,
    cdvsrctli.product_type_id,
    cdvsrctli.quantity, cdvsrctli.final_product_price
FROM cdvsrctli_invoice_items as cdvsrctlii;
```

Dashboard of Product Analysis

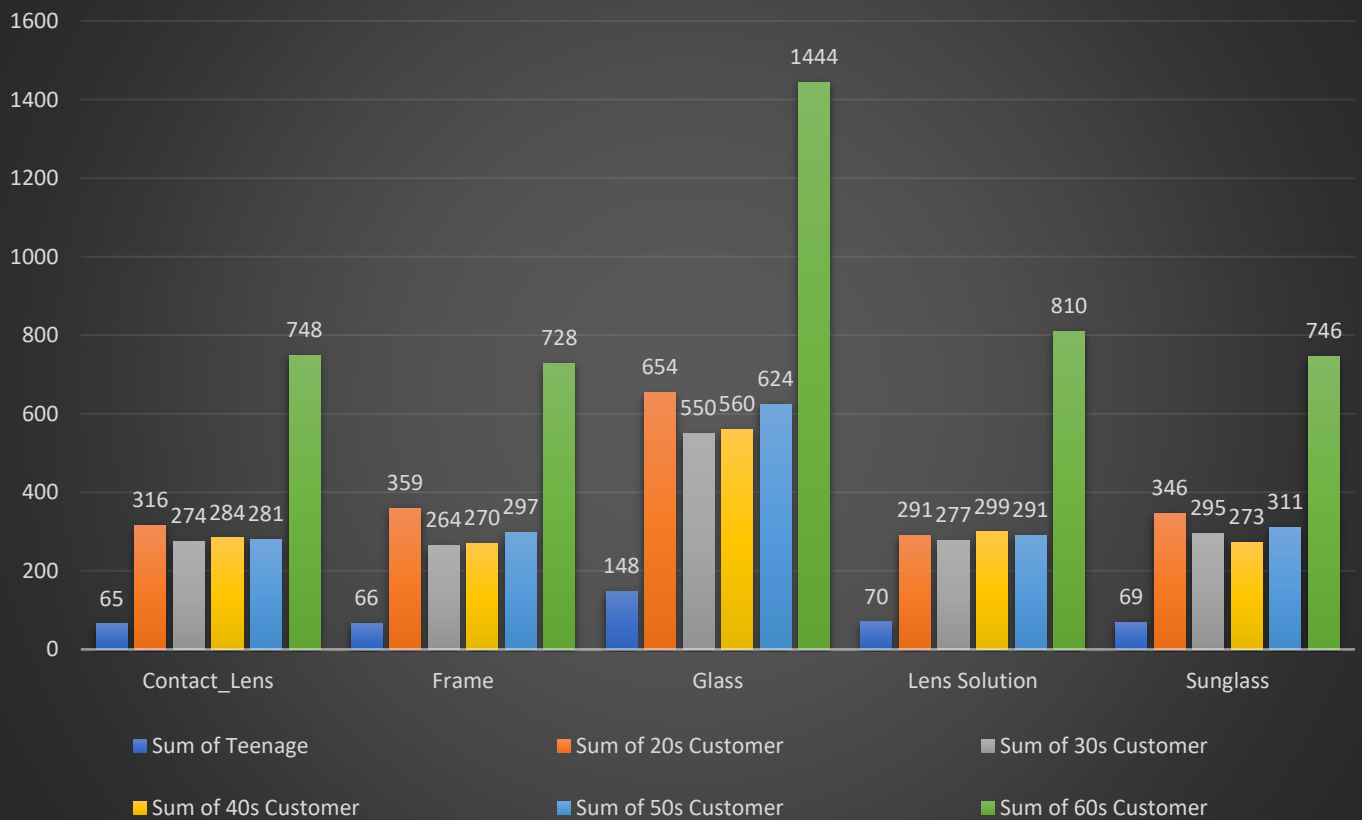
5 Total Products	12,010 Total QTY Sold	4.51 Cr (Rs) Total Revenue
Frames	1984	74.85 L
Glass	3980	1.48Cr
Sunglasses	2040	76.79L
Contact Lens	1968	73.83L
Lens Solution	2038	77.88L



Revenue across Products by Buyers/Repeated/Churn



Product QTY Sold Across Age group



Why Product Analysis Matters

Product and Brand Analysis plays a critical role in retail decision-making as it helps management understand which products and brands drive the highest revenue, sales volume, repeat purchases, and customer demand. These insights enable the business to optimize inventory planning, identify high-margin product categories, strengthen supplier relationships, improve cross-selling opportunities, and understand purchasing preferences across different customer segments. Ultimately, this analysis supports better merchandising decisions while maximizing profitability and reducing inventory-related risks.

Product Analysis

Key Findings & Business Insights

1. Glasses are the Primary Revenue Driver of the Business

The analysis clearly identifies Glasses as the business's flagship product category.

Not only does it record the highest quantity sold (3,980 units), but it also contributes approximately ₹1.48 Crore in revenue, accounting for nearly 33% of the company's total product revenue, almost twice the contribution of every other product category.

This indicates that the business is fundamentally driven by prescription eyewear rather than accessories or secondary products.

Furthermore, cross-analysis with customer demographics shows that Old Age Customers (>60 years) are the largest purchasers of glasses, which aligns perfectly with previous customer segmentation findings where this age group also demonstrated:

- Highest customer volume
- Highest repeat customer base
- Highest revenue contribution

This consistency across multiple analyses validates that senior customers represent the company's core customer segment.

2. Progressive Lenses Present the Largest Cross-Selling Opportunity

Although the project analysis products at a higher level, the findings naturally indicate another important business opportunity.

Senior customers frequently replace their prescription lenses due to changing eye conditions every few years. This creates a recurring opportunity to recommend:

- Progressive lenses
- Premium anti-glare coatings
- Blue-cut lenses

- Premium spectacle frames

Among these, Progressive Lenses deserve special attention because they:

- Generate significantly higher profit margins
- Meet the needs of aging customers
- Increase average bill value
- Improve long-term customer satisfaction

Since senior customers already form the largest repeat customer segment, recommending premium lens upgrades can substantially increase customer lifetime value.

3. Cross-Selling Frames Can Significantly Increase Revenue

The analysis also reveals an overlooked business opportunity.

Many customers purchase only replacement lenses while continuing to use their existing spectacle frames.

This creates an ideal opportunity for staff to recommend new premium frames during every prescription replacement.

Selling a new frame together with prescription lenses can substantially increase average order value while improving customer satisfaction through updated styles and improved comfort.

Considering spectacle frames generally carry retail margins ranging between **100–200%**, increasing frame attachment rates can become one of the most profitable sales strategies for the business.

4. Churn Among Glass Buyers Represents a Significant Revenue Risk

One of the most important findings from the churn analysis is that a considerable number of **Glass buyers** have already entered the churn segment.

These customers collectively represent approximately **₹15 Lakhs** of historical revenue that is now at risk of being permanently lost if re-engagement efforts are not initiated.

Since prescription customers typically require periodic eye examinations and lens replacements, proactive follow-ups through:

- Eye check-up reminders
- WhatsApp campaigns
- Phone calls
- Personalized offers
- Frame upgrade recommendations

We can recover a meaningful portion of this revenue before these customers permanently shift to competitors.

5. Product Portfolio is Well Balanced

Apart from Glasses, the remaining product categories—including:

- Contact Lenses
- Spectacle Frames
- Sunglasses
- Lens Solutions

all contribute relatively similar sales volumes and revenue.

Their contribution differs only marginally, indicating that the business is not overly dependent on any single secondary product category.

This balanced product mix reduces operational risk while ensuring stable revenue generation from multiple product lines.

Product Brand Analysis

5

Total Products

25

Total Brands

201

Total Variety of Products

Ray Ban

Prada

Prime

Bonzer

Renu Fresh

Essilor

Acme

Crizal

Local Brand

Aqua
Soft Bio

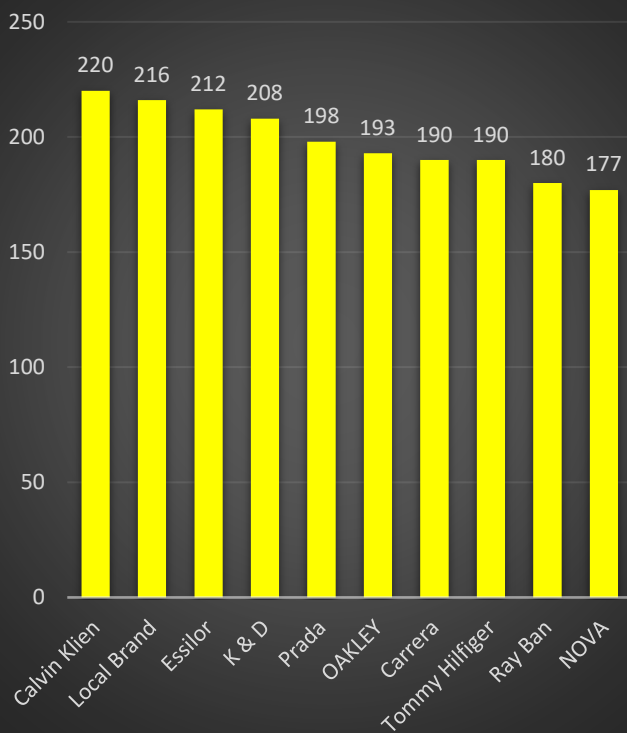
NOVA

Ziess

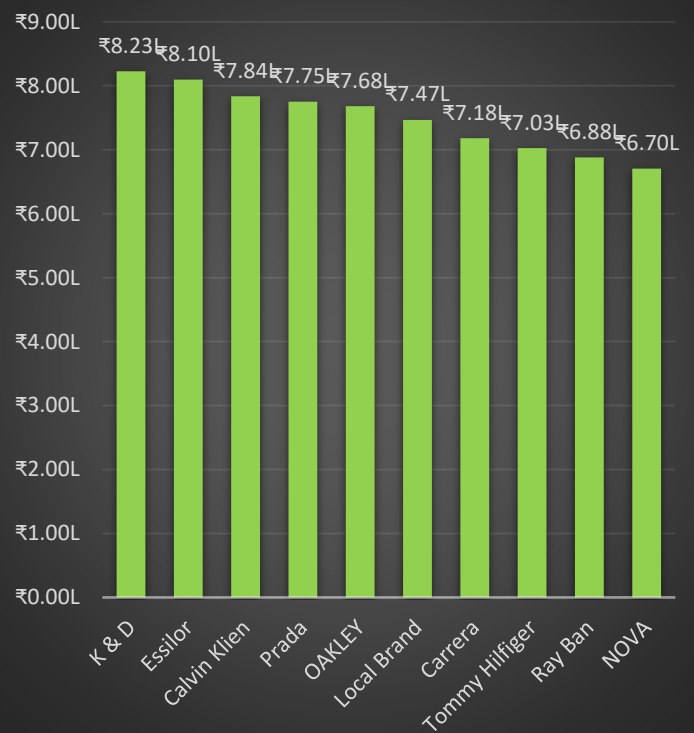
Oakley

Baush & Lomb

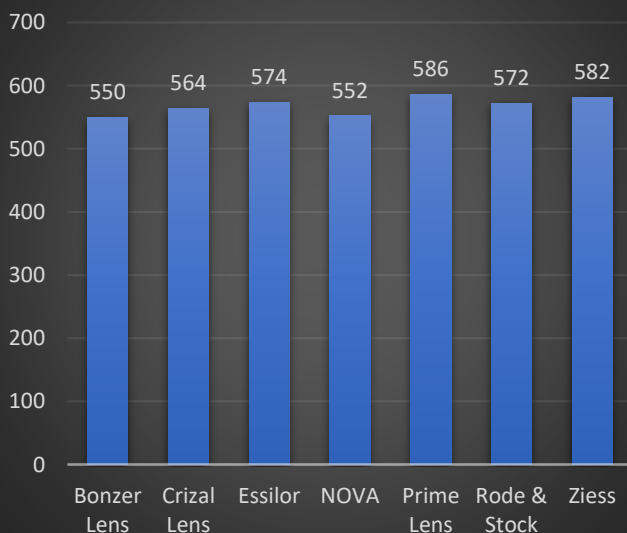
QTY SOLD by each Frame Brand



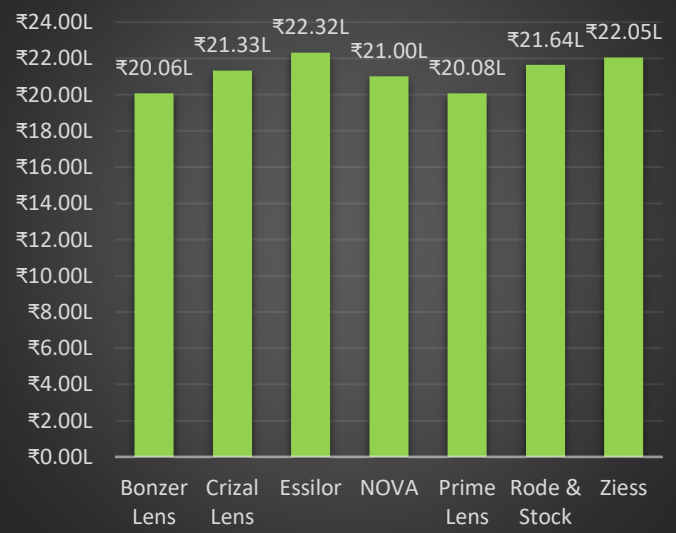
Revenue by each Frame Brand



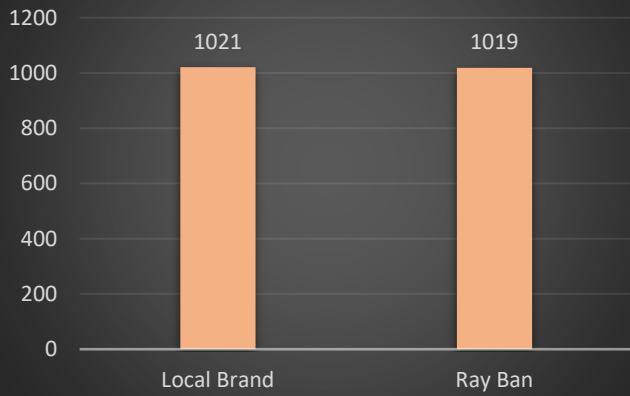
QTY sold by each Glass Brand



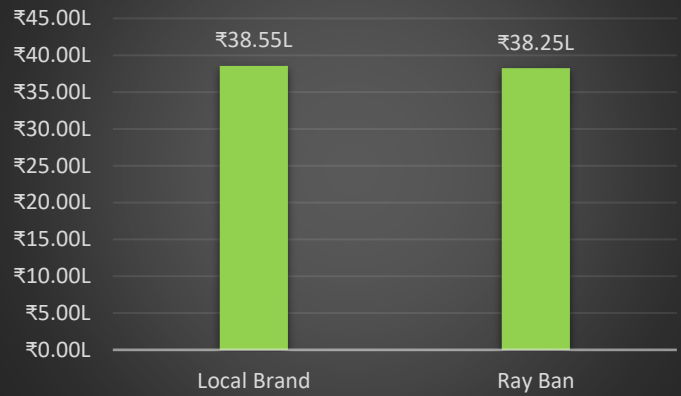
Revenue by each Glass Brand



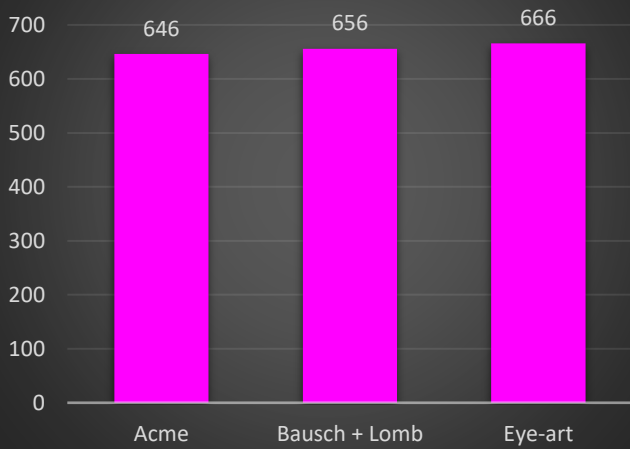
QTY Sold by each Sunglass Brand



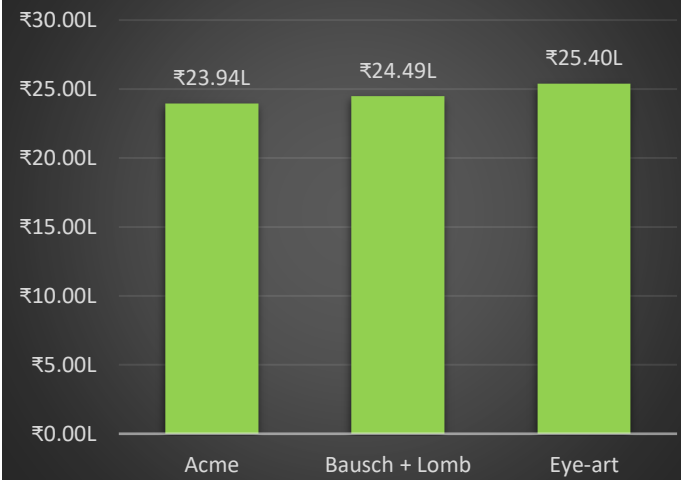
Revenue by each Sunglass Brand



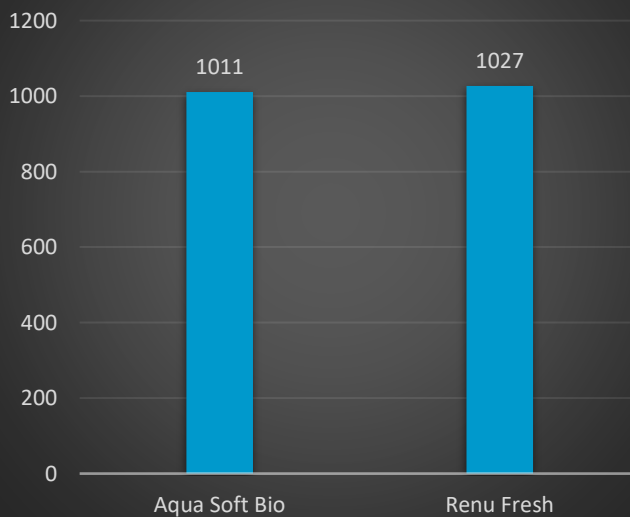
Contact Lens QTY Sold by each Brand



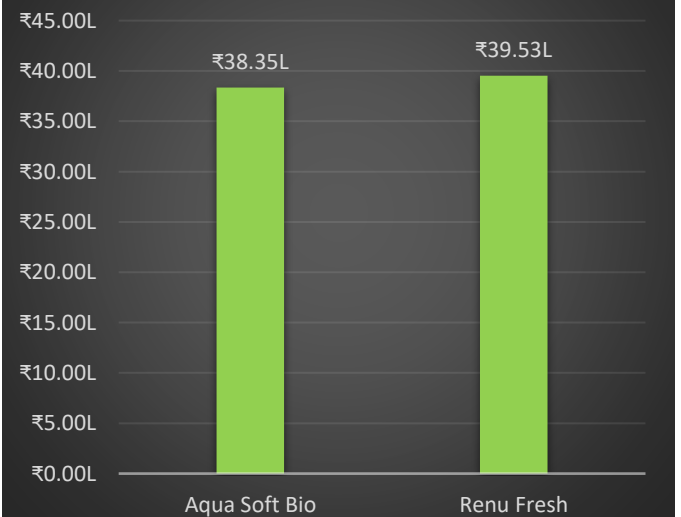
Revenue by each Contact Lens Brand



Lens Solution QTY Sold by each Brand



Revenue by each Lens Solution Brand



Product Brand Analysis

Unlike the product category analysis, the brand-level analysis reveals a different and equally positive business story.

Instead of being dependent upon one dominant brand, the business maintains a healthy and diversified brand portfolio across nearly all product categories.

Key Findings & Business Insights

1. Frame Sales are Distributed Across Multiple Brands

The Frame category demonstrates excellent brand diversification.

Brands such as:

- Calvin Klein
- Essilor
- Local Brand
- KD
- Prada

appear consistently among both:

- Highest quantity sold
- Highest revenue generated

This indicates that customer demand is spread across multiple brands rather than concentrated around a single premium label.

Such diversification reduces supplier dependency while giving customers a wider range of purchasing options across different price segments.

2. Glass Lens Brands Show Balanced Customer Demand

One of the strongest operational findings appears within the Glass category.

Sales quantities across major lens brands remain remarkably similar.

Brands including:

- Essilor
- Crizal
- Prime
- Zeiss

- NOVA
- Bonzer

all contribute comparable sales volumes and revenue.

This indicates that customers trust multiple lens brands rather than demanding only one premium option.

For inventory planning, this is an excellent outcome because inventory investment remains evenly distributed instead of being concentrated into one supplier.

3. Local Sunglasses Successfully Compete Against Ray-Ban

Perhaps the most interesting competitive insight emerges from the Sunglasses category.

The analysis shows that Local Brand Sunglasses perform almost identically to Ray-Ban in both:

- Quantity Sold
- Revenue Generated

This suggests that customers are highly receptive to affordable alternatives when product quality and styling meet expectations.

For management, this provides an opportunity to continue promoting high-margin local brands while maintaining premium international brands for customers seeking branded products.

4. Contact Lens Brands Demonstrate Stable Demand

The Contact Lens category also exhibits a balanced distribution.

Brands such as:

- Acme
- Bausch & Lomb
- Eye-Art

all generate comparable sales quantities and revenue.

This indicates consistent customer acceptance across manufacturers and reduces operational dependence on any single supplier.

5. Lens Solutions Generate Stable Supporting Revenue

Although Lens Cleaning Solutions contribute a smaller share compared to prescription products, they collectively generate approximately **₹80 Lakhs** in revenue.

This is a significant supporting revenue stream.

Because every spectacle and contact lens user requires regular cleaning solutions, this category represents an excellent opportunity for:

- Bundle offers
- Add-on sales
- Checkout recommendations
- Subscription-style replacement reminders

Increasing attachment rates for these products requires minimal selling effort while improving average order value.

6. Brand Portfolio Risk is Well Diversified

Perhaps the strongest overall conclusion from the brand analysis is that:

The business is not overly dependent on any single brand across any product category.

Revenue generation remains well distributed among multiple suppliers, reducing procurement risk while providing customers with greater choice across different pricing segments.

This diversification strengthens inventory stability and improves the business's resilience against supplier-related disruptions.

Overall Product Analysis Conclusion

The product analysis demonstrates that the business possesses a strong and balanced merchandising strategy.

While prescription **Glasses** remain the primary revenue engine driven largely by loyal senior customers the remaining product categories contribute consistently to overall business performance.

Similarly, brand analysis shows healthy diversification across suppliers, reducing operational risk while ensuring broad customer choice.

The most valuable business opportunities identified include:

- Increasing Progressive Lens adoption
- Cross-selling premium spectacle frames
- Recovering churned Glass buyers through proactive follow-ups
- Promoting high-margin local brands alongside premium brands
- Bundling Lens Solutions with eyewear purchases to increase average bill value

Overall, the product portfolio appears both commercially strong and operationally balanced, providing an excellent foundation for sustainable long-term revenue growth.